Does Curiosity Shape Transferable Representations? Intrinsic Motivation as a Representation Learning Mechanism for Cross-Task Transfer in Procgen

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Abstract

Intrinsic motivation methods such as ICM, RND, and CFN are widely used to encourage exploration in reinforcement learning, but their effect on *representation quality* and *cross-task transferability* remains poorly understood. We present a systematic empirical study of four intrinsic reward methods—ICM, RND, CFN, and NLL-Surprise—across 507 trained checkpoints spanning 12 Procgen games, evaluating whether curiosity-driven exploration during source training produces representations that transfer better to related target tasks. We introduce a representational analysis framework comprising linear probes, effective rank, CKA similarity, and Grad-CAM saliency to characterize representation health independently of downstream task performance. Our key findings are: (1) ICM with a midtraining cutoff produces the most transferable representations on the primary Maze→Heist pair, achieving 0.914 action probe accuracy versus 0.812 for vanilla PPO; (2) effective rank is a strong within-pair predictor of transferability ($r = 0.913$), but this correlation does not generalise across game pairs ($r = 0.093$); (3) the intrinsic reward weight η is the critical hyperparameter, with high values causing catastrophic representation collapse; (4) the cutoff mechanism—deactivating intrinsic rewards after early training—functions as a *rescue mechanism* for runs where the intrinsic signal has drifted: it produces dramatic gains on the Maze→Heist pair ($+0.211$ action probe accuracy on Heist) and for cross-environment RND transfer, but is a wash on average across 12 games when always-on ICM is already productive; (5) always-on intrinsic motivation hurts representation quality in the majority of 12 tested source games, with vanilla PPO producing the best representations in 8 of 12; and (6) NLL-Surprise is uniquely fragile, collapsing representations even at moderate η. These results reframe intrinsic motivation from an exploration heuristic to a *scheduled* representation learning intervention, and reveal that the cutoff mechanism—not intrinsic motivation itself—is the key driver of representational benefit.

1 Introduction

A central question in both reinforcement learning (RL) and cognitive science is what enables behaviour to *generalise*. Strong performance on one task does not imply flexible adaptation when goals, dynamics, or rewards change. From a cognitive-science perspective, intelligence is not merely task success but the ability to build reusable internal representations that support transfer across related settings [8].

Curiosity-driven learning is relevant to this question because biological learners do not rely solely on external reward: they also seek novelty, surprise, and predictable structure [10]. The Intrinsic Curiosity Module (ICM) [11] captures this idea by rewarding transitions that are hard to predict in a learned feature space. While ICM and related methods are typically studied as exploration tools for single-task performance, they may also shape representations during source-task training in ways that affect downstream transfer.

This paper asks: **does intrinsic motivation during source training produce representations that transfer better across tasks?** We treat transfer not merely as an engineering benchmark, but as an operational test of representational generality. If curiosity changes what the agent learns about the world, that should manifest in the linear decodability of task-relevant features, the geometric health of the representation space, and ultimately zero-shot and fine-tuned performance on a related target task.

We conduct a systematic empirical study across 507 trained checkpoints spanning 12 Procgen source games, four intrinsic motivation methods (ICM, RND, CFN, NLL-Surprise), four intrinsic reward weights (η), cutoff versus always-on schedules, and multiple random seeds. We develop a representational analysis framework that measures transferability independently of downstream performance, revealing a nuanced picture: **representation health**—quantified by effective rank and linear probe accuracy—is a strong within-pair predictor of transfer success, but does not generalise as a universal cross-game diagnostic. Moreover, always-on intrinsic motivation *hurts* in most games, and it is the **cutoff schedule** that unlocks the representational benefit.

Our main contributions are:

1. A systematic comparison of four intrinsic motivation methods as *representation learning* interventions across 12 Procgen games, revealing that always-on curiosity degrades representations in the majority of settings.
2. Evidence that effective rank predicts transferability within a fixed game pair ($r = 0.913$) but not across game pairs ($r = 0.093$), establishing its scope and limits as a diagnostic.
3. Identification of the **cutoff mechanism**—deactivating intrinsic rewards midtraining—as essential for representation health, with benefits that generalise across source environ-

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ments (Maze, Heist, Miner → Heist).

4. A characterisation of the η sensitivity landscape, showing a phase transition from healthy to collapsed representations that is method-dependent.

2 Related Work

Intrinsic motivation in RL. Intrinsic reward methods incentivise exploration by augmenting the extrinsic reward signal with a bonus derived from prediction error, novelty, or state visitation counts. ICM [11] uses forward-model prediction error in a learned feature space, with an inverse model to filter out action-irrelevant features. RND [3] measures the discrepancy between a fixed random target network and a trained predictor, providing a simpler novelty signal without dynamics modelling. CFN [9] estimates pseudocounts via coin-flip classification accuracy, offering a count-based alternative that is robust to the noisy-TV problem [4]. NLL-Surprise [1] uses the negative log-likelihood of a Gaussian dynamics model, capturing aleatoric uncertainty through learned variance.

Transfer learning in RL. Transfer learning seeks to leverage knowledge from a source task to accelerate learning or improve performance on a target task [17, 18]. In the Procgen benchmark [5], agents must generalise across procedurally generated levels that share visual style and action space but differ in layout and dynamics. Prior work has studied transfer via progressive neural networks [13], policy distillation [14], and representation learning [16]. However, the effect of the *exploration strategy* during source training on downstream transferability has received limited attention.

Representation analysis. Linear probing [2] is a standard diagnostic for representation quality: a linear classifier trained on frozen features measures how much task-relevant information is linearly accessible. Effective rank [12], computed from the entropy of normalised singular values, quantifies the dimensionality utilisation of a representation. Centered Kernel Alignment (CKA) [7] compares representational geometry across networks by measuring the similarity of their kernel matrices.

3 Methods

3.1 Agent Architecture

All agents use an IMPALA-style CNN encoder [6] followed by a linear projection and actor-critic heads. The encoder consists of three convolutional sequences, each containing a convolution, max-pooling, and two residual blocks with ReLU activations. The output is flattened and projected to a 256-dimensional hidden representation $\mathbf{h} \in \mathbb{R}^{256}$, from which the actor head produces logits over 15 discrete actions and the critic head produces a scalar value estimate. Policy optimisation uses PPO [15] with standard hyperparameters: learning rate 5×10^{-4}, discount factor $\gamma = 0.999$, GAE parameter $\lambda = 0.95$, clip coefficient $\epsilon = 0.2$, entropy coefficient 0.01, and value function coefficient 0.5.

3.2 Intrinsic Motivation Methods

All intrinsic reward methods share a unified interface: given a transition (s_t, a_t, s_{t+1}), each method computes a scalar intrinsic reward r_t^i. The total reward is

$$r_t = r_t^e + \eta \cdot \tilde{r}_t^i, \tag{1}$$

where r_t^e is the extrinsic reward, $\tilde{r}_t^i$ is the RMS-normalised intrinsic reward, and $\eta \in \{0.001, 0.005, 0.01, 0.05\}$ controls the intrinsic reward strength. RMS normalisation divides r_t^i by a running estimate of $\sqrt{\mathrm{Var}(r^i)}$, ensuring that the intrinsic signal remains at unit scale regardless of the method.

3.2.1 ICM (Intrinsic Curiosity Module)

ICM [11] learns a feature encoder ϕ, an inverse dynamics model g, and a forward dynamics model f. The intrinsic reward is the forward prediction error:

$$r_t^i = \|\hat{\phi}(s_{t+1}) - \phi(s_{t+1})\|_2^2, \tag{2}$$

where $\hat{\phi}(s_{t+1}) = f(\phi(s_t), a_t)$. The training loss is $\mathcal{L} = \beta\mathcal{L}_{\text{forward}} + (1-\beta)\mathcal{L}_{\text{inverse}}$, with $\beta = 0.2$. The inverse model trains the encoder to capture action-relevant features, while the forward model provides the novelty signal.

3.2.2 RND (Random Network Distillation)

RND [3] uses a fixed, randomly initialised target network f_{target} and a trainable predictor f_{pred}:

$$r_t^i = \|f_{\text{pred}}(s_t) - f_{\text{target}}(s_t)\|_2^2. \tag{3}$$

Novel states yield high prediction error because the predictor has not been trained on similar inputs. Unlike ICM, RND requires no dynamics model and depends only on observation novelty.

3.2.3 CFN (Coin Flip Networks)

CFN [9] estimates pseudocounts via $K = 32$ independent binary classifiers. For each state, random binary labels are generated by projecting features through a fixed random matrix, and classifier accuracy serves as a proxy for visitation count:

$$\hat{N}(s) = \frac{\bar{a}^2}{(1-\bar{a})^2 + \varepsilon}, \quad r_t^i = \frac{1}{\sqrt{\hat{N}(s_t)}}, \tag{4}$$

where $\bar{a}$ is the mean classification accuracy across the K heads. Frequently visited states have high accuracy (high pseudocount) and low reward.

3.2.4 NLL-Surprise

NLL-Surprise [1] models transitions with a Gaussian dynamics model that outputs mean $\boldsymbol{\mu}$ and log-variance $\log \boldsymbol{\sigma}^2$ for the next-state features:

$$r_t^i = -\log p_\theta(\phi(s_{t+1}) \mid \phi(s_t), a_t) \tag{5}$$

$$= \tfrac{1}{2} \sum_d \left(\log \sigma_d^2 + \frac{(\phi_d - \mu_d)^2}{\sigma_d^2} \right) + C. \tag{6}$$

Unlike ICM's MSE-based error, NLL captures aleatoric uncertainty through the learned variance, which in principle allows the model to downweight inherently stochastic transitions.

3.3 Cutoff Mechanism

Procedurally generated environments pose a unique challenge for intrinsic motivation: because every episode presents novel level layouts, the curiosity signal never naturally decays. We introduce a **cutoff schedule** that deactivates all intrinsic rewards after a fixed number of training steps (10M of 25M total), creating an explore-then-exploit curriculum. After the cutoff, only extrinsic reward drives policy optimisation, allowing the agent to consolidate task-relevant features without the distraction of persistent novelty bonuses.

3.4 Environments

We use the Procgen benchmark [5], which provides 16 procedurally generated games sharing a common visual style (64×64 RGB) and discrete action space (15 actions). Our primary transfer pair is **Maze** (source) → **Heist** (target): Maze emphasises navigation through corridors, while Heist requires key collection and door interaction, testing whether navigation-oriented representations generalise to object-interaction tasks. We additionally evaluate cross-environment transfer from Heist and Miner as source environments. To test the generality of our findings beyond a single game pair, we conduct a broad 12-game study spanning Bigfish, Chaser, Climber, Coinrun, Dodgeball, Heist, Jumper, Leaper, Maze, Ninja, Plunder, and Starpilot, probing each game's agents against its own expert. All experiments use `easy` distribution mode, 64 parallel environments, and 200 training levels, with the exception of the cross-environment RND experiments (Maze/Heist/Miner → Heist, Section 6.8) which use `hard` distribution mode.

4 Evaluation Framework

4.1 Transfer Protocols

We evaluate transfer through three protocols:

1. **Zero-shot**: All weights frozen; evaluated on 2000 Heist episodes with deterministic (argmax) actions.
2. **Fresh-head finetune**: Encoder frozen; actor and critic heads reinitialised and trained on Heist for 5M steps with PPO.
3. **Unfreeze-1 finetune**: Encoder frozen; existing heads unfrozen with a critic-only warmup (10 iterations), then 5M steps on Heist.

Results are reported as mean reward with win rate and standard deviation across 3 seeds. We also train a **from-scratch baseline**: vanilla PPO initialised randomly and trained directly on Heist for 5M steps.

4.2 Representational Analysis

To characterise representation quality independently of downstream task performance, we develop a suite of representational diagnostics applied to frozen source-task encoders:

Linear probes. We train single-layer linear classifiers on frozen 256-dimensional encoder features to predict (a) the *action probe*: the expert Heist agent's argmax action across 15 classes, and (b) the *value probe*: the expert's value function $V(s)$. The expert dataset consists of 200 episodes on held-out Heist levels (levels 500+, easy mode). Probes are trained for 200 epochs with Adam (lr $= 10^{-3}$, weight decay 10^{-4}) on an 80/20 train/test split. The action probe reports test accuracy; the value probe reports $R^2 = 1 - \mathrm{MSE}/\mathrm{Var}(y)$.

High action probe accuracy indicates that the encoder already represents Heist states in a way that supports linear readout of the correct action—a direct measure of feature transferability. High value R^2 indicates that state-value information is linearly accessible, reflecting whether the representation encodes task-relevant state quality.

Effective rank. The encoder produces a 256-dimensional feature vector $\mathbf{h} \in \mathbb{R}^{256}$ for each observation, but not all dimensions necessarily carry useful information: some may always output zero (dead neurons), while others may be highly correlated (redundant). Effective rank quantifies how many dimensions are *actually used*.

Given the $N \times 256$ feature matrix $\mathbf{X}$ (observations × features), we compute the singular values $\sigma_1 \geq \cdots \geq \sigma_{256}$, normalise them to form a distribution $p_i = \sigma_i / \sum_j \sigma_j$, and define the effective rank as $\exp(H(\mathbf{p}))$, where H is Shannon entropy [12]. If all 256 dimensions carry equal variance, the effective rank is 256 (full capacity utilisation). If variance concentrates on a single dimension, the effective rank is 1 (collapsed). In our experiments, healthy representations (e.g. ICM cutoff) achieve an effective rank of ∼70, meaning roughly 70 of 256 dimensions carry meaningful, independent information. Collapsed representations (e.g. NLL at high η) fall to an effective rank of ∼4, with over 85% of neurons completely dead—the encoder has discarded almost all information about the observation. We also report *dead neuron fraction*: the proportion of features with zero activation across all observations, which provides a complementary measure of representation health.

CKA (Centered Kernel Alignment). For two feature matrices $\mathbf{X}, \mathbf{Y} \in \mathbb{R}^{N \times D}$ computed from the same observations, linear CKA [7] measures representational similarity:

$$\mathrm{CKA}(\mathbf{X}, \mathbf{Y}) = \frac{\|\mathbf{Y}^\top \mathbf{X}\|_F^2}{\|\mathbf{X}^\top \mathbf{X}\|_F \cdot \|\mathbf{Y}^\top \mathbf{Y}\|_F}, \tag{7}$$

after mean-centering both matrices. CKA $= 1$ indicates identical representational geometry (up to linear transform); CKA $= 0$ indicates orthogonal representations. We compute pairwise CKA across all checkpoints to reveal method-family clustering and identify collapsed representations.

5 Experimental Setup

5.1 Training Configuration

Source-task agents are trained on Maze for 25M timesteps (64 parallel environments, 256-step rollouts). Intrinsic reward methods use a separate optimiser (Adam, lr $= 10^{-3}$) with the same observation stream. Cutoff variants deactivate intrinsic rewards and freeze the intrinsic model after 10M steps.

5.2 Experiment Phases

We conduct three experiment phases totalling approximately 700 GPU-hours on NVIDIA RTX 3090 and A100 GPUs:

1. **Eta sweep** (single seed): 20 conditions—4 methods $\times$ 4 η values, plus cutoff variants (ICM, RND, NLL at $\eta = 0.01$) and a vanilla baseline—to identify the best η per method.
2. **NLL rerun** (single seed): Verification of NLL-Surprise results after observing extreme variance between runs.
3. **Multi-seed sweep** (3 seeds): 10 best conditions $\times$ 3 seeds = 30 runs to establish statistical reliability.

5.3 Representational Analysis Configuration

The representational analysis pipeline processes checkpoints against expert datasets, extracting 256-dimensional features for action and value probes, effective rank, dead neuron statistics, and pairwise CKA. Data sources span five experiment sets totalling 507 checkpoints:

- **Primary** (Fawnia): 9 method variants $\times$ 3 seeds, Maze $\rightarrow$ Heist, easy distribution mode, probed against a shared Heist expert (200 episodes, held-out levels 500+).
- **Eta sweep** (Yu): 4 methods $\times$ 4 η values $\times$ 3 seeds + cutoffs = 59 checkpoints, easy distribution mode, probed against Heist expert.
- **ICM ablation** (Aggarwal): ICM β/η sweep with 2 seeds each, easy distribution mode, probed against Heist expert.
- **Cross-environment** (Macias): RND from Maze/Heist/Miner $\rightarrow$ Heist (49 checkpoints, always-on and cutoff variants, including sequential training). This is the *only* experiment set using `hard` distribution mode, making cross-mode comparisons with the other sets approximate.
- **12-game survey**: 258 checkpoints across 12 Procgen games $\times$ {ICM, RND, CFN, vanilla} $\times$ $\eta \in \{0.005, 0.01\}$ $\times$ 3–5 seeds, easy distribution mode, each game probed against its own expert. This tests whether intrinsic motivation benefits generalise beyond the Maze$\rightarrow$Heist pair.

6 Results

6.1 Representational Quality by Method

What "representational quality" means here. We operationalise representational quality through three complementary metrics, each probing a different aspect of whether the frozen encoder has learned something useful for the target task:

- **Action probe accuracy** (top panel, Figure 1) asks: *can a linear classifier read the expert's action choice directly from the features?* This measures whether task-relevant *behavioural* information is linearly accessible without any further adaptation. A value of 1.0 means the encoder perfectly separates states that require different actions; chance is $1/15 \approx 0.067$.
- **Value probe** R^2 (middle panel) asks: *can a linear regressor recover the expert's value estimate $V(s)$ from the features?* This measures whether the encoder captures *state quality*—distinguishing promising from unpromising states. $R^2 = 1$ indicates perfect recovery; $R^2 = 0$ indicates no information beyond the mean; negative R^2 indicates the features actively mislead the regressor.
- **Effective rank** (bottom panel) asks: *how many of the 256 feature dimensions carry independent information?* This measures the encoder's dimensional *capacity utilisation*. Low effective rank indicates representation collapse (redundant or dead dimensions); high effective rank indicates distributed, high-capacity features.

Table 1: Representational analysis: linear probe accuracy, value R^2, effective rank, and dead neuron fraction for frozen Maze encoders probed on Heist expert labels. Mean $\pm$ std across 3 seeds. Best per column **bolded**.

Method	Action Acc.	Value R^2	Eff. Rank	Dead %
Vanilla	$.812 \pm .125$	$.500 \pm .214$	53.9 ± 18.8	50.9 ± 16.6
ICM	$.825 \pm .039$	$.520 \pm .035$	62.7 ± 8.9	42.7 ± 10.8
ICM cutoff	$\mathbf{.914 \pm .023}$	$\mathbf{.623 \pm .066}$	$\mathbf{70.6 \pm 2.9}$	$\mathbf{33.7 \pm 5.9}$
RND	$.828 \pm .064$	$.570 \pm .093$	66.6 ± 7.3	41.0 ± 7.5
RND cutoff	$.846 \pm .087$	$.547 \pm .145$	59.7 ± 9.6	45.3 ± 8.8
CFN	$.856 \pm .055$	$.551 \pm .044$	57.4 ± 11.5	47.5 ± 10.8
CFN cutoff	$.855 \pm .086$	$.621 \pm .120$	59.7 ± 13.1	47.7 ± 15.2
NLL	$.204 \pm .078$	$.073 \pm .103$	4.1 ± 4.6	85.7 ± 13.2
NLL cutoff	$.419 \pm .324$	$-.494 \pm .855$	13.2 ± 12.0	66.5 ± 37.0

Together, these three metrics distinguish several failure modes: an encoder can have high rank but low probe accuracy (high-dimensional but uninformative, as with CFN), or high probe accuracy but low rank (narrow but task-aligned), or both collapse together (NLL). A representation is "high quality" only when all three are high simultaneously: the features carry independent information (high rank) that is linearly organised around task-relevant structure (high action accuracy and value R^2). This multi-metric view is necessary because no single metric captures transferability on its own—an encoder that scores 90% action accuracy on a nearly one-dimensional feature space has clearly overfit to a narrow behavioural signal, while one that achieves maximal rank but near-chance probe accuracy has diverse but meaningless features. The importance of using all three is evident in our results: CFN maintains high effective rank across games but has low probe accuracy, while NLL collapses on all three axes simultaneously.

Figure 1 presents the primary method comparison from the multi-seed Maze $\rightarrow$ Heist experiments (3 seeds each). Table **??** provides exact statistics.

ICM with cutoff is the clear winner. ICM cutoff achieves the highest action probe accuracy (0.914 ± 0.023), value R^2 (0.623 ± 0.066), and effective rank (70.6 ± 2.9), with the lowest dead neuron fraction (33.7%). This represents a 12.6 percentage-point improvement in action accuracy over vanilla PPO (0.812 ± 0.125) with substantially reduced variance, indicating both better and more consistent representations.

All methods except NLL improve upon or match vanilla. ICM, RND, and CFN (both always-on and cutoff variants) achieve action probe accuracies in the 0.825–0.856 range, all modestly above vanilla. The cutoff variants show no systematic advantage for RND and CFN at this η value.

NLL-Surprise causes catastrophic representation col-

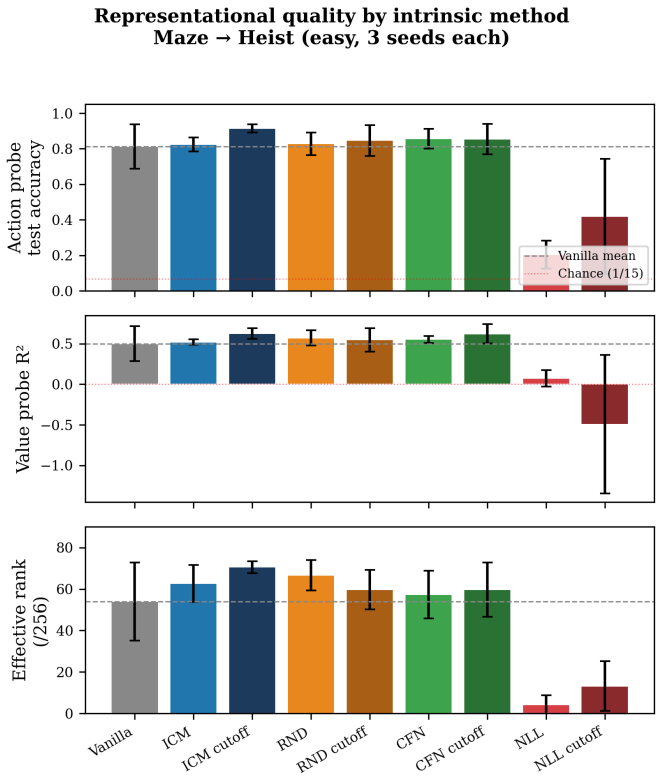

Figure 1: Representational quality by intrinsic method (Maze → Heist, 3 seeds). **Top**: Action probe test accuracy (15-class linear classifier on frozen features). **Middle**: Value probe R^2 (linear regression on expert $V(s)$). **Bottom**: Effective rank of the 256-dimensional representation. Dashed grey line: vanilla PPO baseline. ICM with cutoff dominates all three metrics. NLL-Surprise collapses catastrophically.

lapse. NLL always-on achieves only 0.204 ± 0.078 action accuracy—barely above chance ($1/15 \approx 0.067$)—with an effective rank of 4.1 (out of 256) and 85.7% dead neurons. Even with cutoff, recovery is limited: 0.419 ± 0.324 accuracy with extremely high variance, indicating that the damage from NLL's training signal persists even after deactivation.

Why a single metric is not sufficient: the CFN counter-example. The multi-metric view is not just conceptually cleaner—it is empirically necessary. Consider CFN in the broader 12-game survey (Section 6.7): CFN maintains high effective rank (57–80 across games, often exceeding vanilla) yet achieves the worst action probe accuracy of any method tested (0.505 mean, vs. 0.771 for vanilla). In other words, CFN spreads its representational capacity across many dimensions, but those dimensions do not linearly encode the expert's action choices. This is the opposite failure mode from NLL: NLL collapses to a useless one-dimensional representation, while CFN maintains a high-dimensional *but unstructured* representation. Neither probe accuracy nor effective rank alone would flag both of these failures. Only the combination—high rank *paired with* high probe accuracy and high value R^2—corresponds to representations we observe to transfer well.

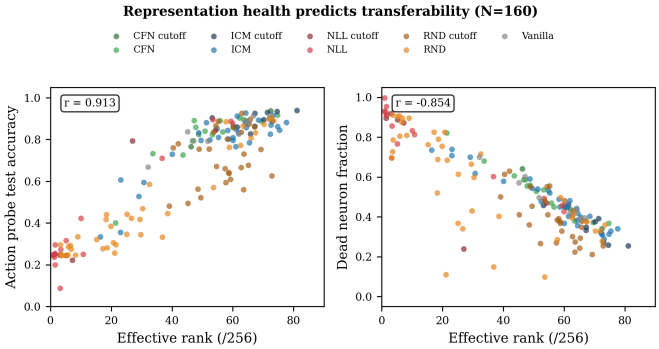

Figure 2: Representation health predicts transferability ($N = 160$ checkpoints). **Left**: Effective rank vs. action probe accuracy ($r = 0.913$). **Right**: Effective rank vs. dead neuron fraction ($r = -0.854$). The NLL-Surprise cluster (bottom-left/top-right) represents collapsed representations; healthy methods form a continuum in the upper-right/lower-left.

6.2 Effective Rank Predicts Transferability

Figure 2 shows the relationship between effective rank and action probe accuracy across all 160 checkpoints pooled from four experiment sets.

The correlation between effective rank and action probe accuracy is strikingly strong ($r = 0.913$, $p < 10^{-10}$), establishing effective rank as a reliable predictor of transferability. This relationship is near-linear across the full range of representation health, from collapsed NLL representations (rank < 10, accuracy < 0.2) to the healthiest ICM cutoff representations (rank > 70, accuracy > 0.9).

The negative correlation with dead neuron fraction ($r = -0.854$) confirms a consistent failure mode: representations with many dead neurons have low effective rank and poor transferability. This suggests that **effective rank can serve as a practical, target-task-independent diagnostic for transfer readiness**—one can evaluate representation health without needing expert labels or a trained target-task agent.

6.3 Variance Concentration: A Complementary Collapse Diagnostic

Effective rank summarises the distribution of variance across all 256 dimensions, but examining the *top principal components* provides a sharper view of collapse severity. Table 2 reports three complementary statistics: top-1 variance fraction (share of total variance in the dominant dimension), top-10 variance fraction, and k_{95} (number of dimensions required to capture 95% of total variance).

NLL collapse is extreme. NLL-always-on concentrates 95.5% of total feature variance in a single dimension, with $k_{95} = 1.5$—the representation is essentially one-dimensional. NLL cutoff recovers modestly to $k_{95} = 5.3$, confirming partial but incomplete recovery.

Healthy methods distribute variance across $\sim$ 30–40 dimensions. All non-NLL methods have k_{95} in the range 32–40, indicating that roughly 30–40 independent feature directions capture most of the representational information. ICM cutoff leads with $k_{95} = 40.3$, consistent with its higher effective rank. This suggests a rough "threshold of representational

Table 2: Variance concentration statistics (Maze→Heist, 3 seeds). **Top-1**: variance fraction in the largest PC. **Top-10**: fraction in the first 10 PCs. k_{95}: dimensions needed for 95% variance. Healthy reps have low Top-1 and high k_{95}.

Method	Top-1	Top-10	k_{95}
ICM cutoff	0.223	0.701	**40.3**
RND	0.247	0.715	38.7
ICM	0.176	0.713	36.3
CFN cutoff	0.177	0.722	35.3
RND cutoff	0.193	0.704	35.3
CFN	0.207	0.753	32.3
Vanilla	0.210	0.726	32.3
NLL cutoff	0.833	0.982	5.3
NLL	**0.955**	**0.998**	**1.5**

utility": representations with $k_{95} < 10$ are collapsed and untransferable, while $k_{95} > 30$ corresponds to useful transfer quality.

6.4 Representational Similarity Structure

Figure 3 shows the pairwise CKA similarity matrix across 59 checkpoints from the eta sweep.

Three structural patterns emerge:

Method-family clustering. Each intrinsic method (CFN, ICM, RND) forms a visible block of elevated CKA ($\sim$ 0.5–0.7) along the diagonal, indicating that same-method agents converge on similar representational geometry despite different η values and random seeds. This suggests that the training signal, not stochastic variation, is the primary determinant of representation structure.

NLL is an outlier. The NLL block shows near-zero CKA with all other methods, confirming that its collapsed representations occupy a fundamentally different region of representation space. This is not merely poor quality—it is a qualitatively different representation.

Cross-method similarity. ICM, RND, CFN, and vanilla share moderate CKA ($\sim$ 0.3–0.5), suggesting partial convergence on overlapping feature spaces. At low η, intrinsic methods produce representations close to vanilla, consistent with the intrinsic signal being a weak perturbation.

6.5 ICM Hyperparameter Sensitivity

Figure 4 presents the β/η ablation for ICM, isolating the effect of the forward/inverse loss balance and intrinsic reward weight.

η is the critical hyperparameter. Within the moderate range ($\eta \in [0.001, 0.005]$), all β values from 0.05 to 0.8 yield action accuracy in the 0.86–0.90 range, with value R^2 of 0.78–0.83 and effective rank of 60–72. This remarkable robustness to β indicates that the balance between forward and inverse model losses is not a sensitive design choice.

$\eta = 0.05$ causes collapse. At this value, action accuracy drops to $\sim$ 0.60, value R^2 falls to $\sim$ 0.40, and effective rank crashes to $\sim$ 27—well below vanilla. This is the "too much curiosity" failure mode: the intrinsic signal dominates extrinsic reward, driving the agent toward surprising but task-irrelevant transitions.

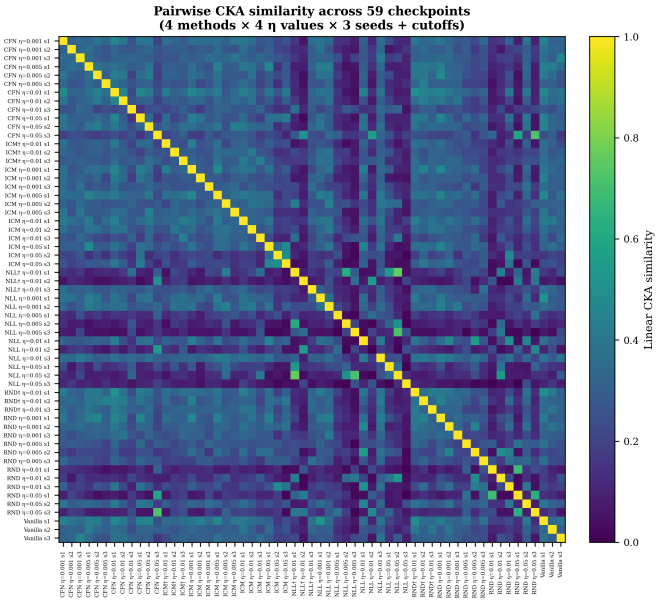

Figure 3: Pairwise CKA similarity across 59 checkpoints (4 methods $\times$ 4 η values $\times$ 3 seeds + cutoff variants + vanilla baselines). Block-diagonal structure indicates that same-method agents learn more similar representations regardless of η or seed. NLL rows/columns are notably dark, confirming that collapsed representations are structurally dissimilar to all healthy representations.

Cutoff rescues moderate-η runs. ICM cutoff at $\eta = 0.005$ achieves the best overall result ($\sim$ 0.93 action accuracy, $\sim$ 75 effective rank), exceeding both always-on ICM and vanilla by a substantial margin. The cutoff allows the agent to benefit from early curiosity-driven exploration while consolidating on task reward during later training.

6.6 Effect of Intrinsic Reward Strength

Figure 5 characterises the η sensitivity landscape across all four methods using William's sweep data (3 seeds, always-on, no cutoff).

Phase transition around $\eta \in [0.01, 0.05]$. All methods perform at or above vanilla at low η (≤ 0.005), but a dramatic divergence occurs at higher values. NLL collapses completely at $\eta = 0.05$ (near-zero accuracy, near-zero rank, $\sim$ 100% dead neurons). RND also degrades substantially. CFN shows moderate sensitivity.

ICM is the most η-robust method. ICM maintains reasonable representation quality even at $\eta = 0.05$, likely because its inverse model constrains the feature encoder to capture action-relevant information, preventing the unconstrained feature drift that collapses NLL and RND representations.

The collapse is abrupt, not gradual. The transition from healthy to collapsed representations occurs over less than one order of magnitude in η, suggesting a bifurcation in the optimisation landscape rather than a smooth degradation.

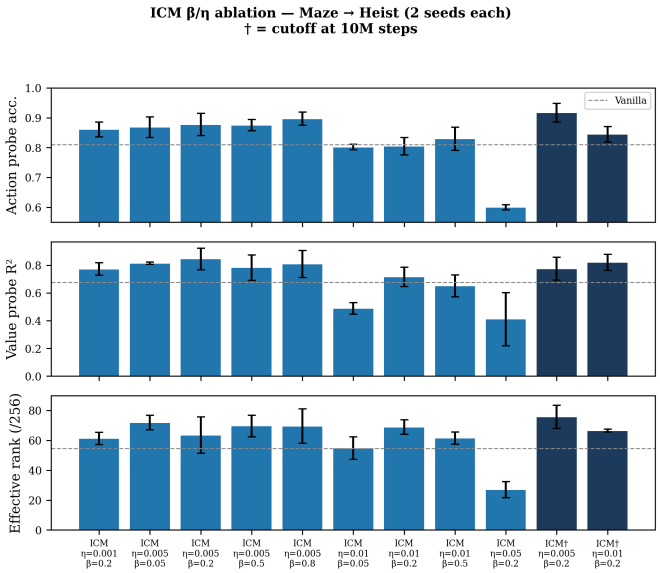

Figure 4: ICM β/η ablation (Maze $\rightarrow$ Heist, 2 seeds each). † indicates cutoff at 10M steps. Dashed line: vanilla baseline. β (forward/inverse balance) has minimal effect within the moderate-η regime, while $\eta = 0.05$ causes sharp degradation. ICM† with $\eta = 0.005$ achieves the best overall action accuracy (~ 0.93).

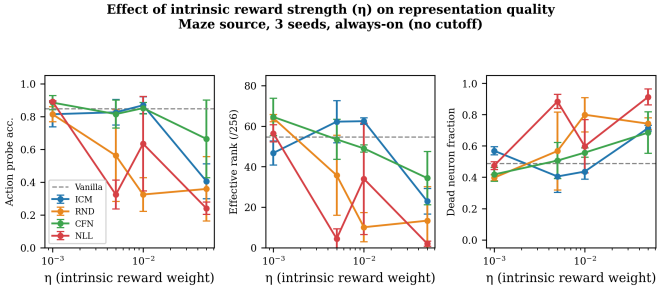

Figure 5: Effect of intrinsic reward strength (η) on representation quality (Maze source, 3 seeds, always-on). **Left**: Action probe accuracy. **Center**: Effective rank. **Right**: Dead neuron fraction. A phase transition occurs around $\eta \in [0.01, 0.05]$: NLL and RND collapse while ICM remains relatively robust. Dashed line: vanilla baseline.

6.7 Broad Game Survey: Always-On Curiosity Usually Hurts

To test whether our Maze$\rightarrow$Heist findings generalise, we evaluate 258 always-on checkpoints across 12 Procgen games, each probed against its own expert. Table 3 summarises action probe accuracy by game and method.

Vanilla is the best method in 8 of 12 games. When intrinsic reward is left on for the full 25M steps, it degrades representations more often than it helps. The mean action accuracy across all games is highest for vanilla (0.771), followed by ICM (0.757), RND (0.636), and CFN (0.505).

CFN consistently hurts. CFN underperforms vanilla in all 12 games, with a mean deficit of -0.266. Despite maintaining high effective rank (often > 75), CFN produces representations with poor linear decodability, further confirming that high rank alone is not sufficient for transfer quality.

Table 3: Action probe accuracy across 12 Procgen games (mean over 3–5 seeds, always-on, $\eta \in \{0.005, 0.01\}$). Each game is probed against its own expert. Best method per game **bolded**. Vanilla is best in 8 of 12 games.

Game	Vanilla	ICM	RND	CFN
Plunder	**0.935**	0.905	0.639	0.582
Bigfish	**0.905**	0.889	0.733	0.563
Starpilot	**0.901**	0.868	0.827	0.517
Chaser	0.899	**0.914**	0.878	0.553
Climber	**0.890**	0.793	0.613	0.542
Dodgeball	0.865	**0.875**	0.821	0.567
Heist	**0.807**	0.660	0.345	0.535
Ninja	**0.732**	0.557	0.518	0.409
Jumper	**0.626**	0.550	0.468	0.382
Maze	0.599	**0.869**	0.537	0.556
Leaper	0.585	0.563	**0.805**	0.377
Coinrun	0.512	**0.636**	0.445	0.477
Mean	**0.771**	0.757	0.636	0.505

ICM helps selectively on hard difficulty games. ICM's largest gains occur on games where vanilla representations are weakest: Maze ($+0.270$) and Coinrun ($+0.124$). On games where vanilla already achieves > 0.85 accuracy, ICM offers no improvement or slightly degrades performance. Quantitatively, the correlation between vanilla's action accuracy and ICM's advantage over vanilla is $r = -0.411$ across the 12 games, indicating that ICM helps more when the baseline representation is weaker. The same pattern holds even more strongly for CFN ($r = -0.878$) and RND ($r = -0.414$): intrinsic motivation adds useful representation diversity when the baseline representation is under-specified, but introduces noise when the baseline already captures the relevant features.

The effective rank correlation does not generalise across games. Within the Maze$\rightarrow$Heist pair, effective rank predicts action accuracy with $r = 0.913$ (Section 6.2). Across all 258 checkpoints in the 12-game survey, this correlation drops to $r = 0.093$—essentially zero. This is because different games have different intrinsic difficulty for linear probing: a high-rank Coinrun representation (rank ~ 65) still achieves lower accuracy than a low-rank Plunder representation (rank ~ 80), because the tasks differ in complexity. Effective rank remains a reliable diagnostic *within* a fixed game pair, but it cannot be compared across game pairs as an absolute measure of transferability.

6.8 Cross-Environment Transfer and the Universality of Cutoff

Figure 6 evaluates cross-environment transfer using RND agents trained on Maze, Heist, and Miner, all probed against the Heist expert dataset. This experiment tests whether the cutoff benefit observed on Maze$\rightarrow$Heist generalises across source environments.

Cutoff is essential across all source environments. Table 4 quantifies the cutoff effect. In every source environment, the cutoff variant approximately doubles action probe accuracy and effective rank while halving dead neuron fraction.

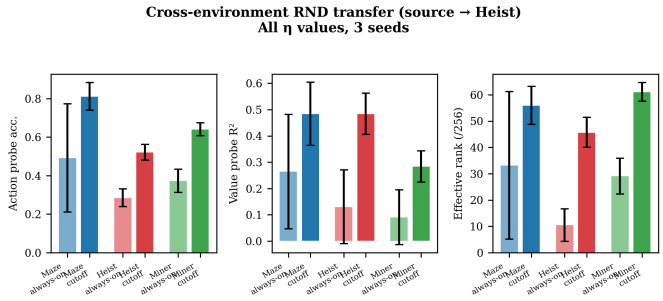

Figure 6: Cross-environment RND transfer (source → Heist). All η values, 3 seeds. Cutoff variants (dark bars) consistently outperform always-on (light bars) across all source environments. Maze with cutoff is the best source ($\sim$ 0.81 action accuracy); Miner with cutoff is also competitive ($\sim$ 0.63), demonstrating viable cross-environment transfer.

Table 4: RND cutoff effect across source environments (all → Heist). Mean across η values and 3 seeds. Cutoff at 10M of 25M steps.

Source	Variant	Action Acc.	Eff. Rank	Dead %
Maze	Always-on	0.493	33.2	59.0%
	Cutoff	**0.811**	**56.0**	**44.1%**
Heist	Always-on	0.285	10.5	87.4%
	Cutoff	**0.521**	**45.7**	**43.8%**
Miner	Always-on	0.373	29.1	37.6%
	Cutoff	**0.641**	**61.1**	**31.2%**

This is the most consistent finding across all experiments: the persistent intrinsic signal degrades representations regardless of the source task.

Source environment matters. Maze with cutoff achieves the highest action accuracy (0.811) and the most task-relevant features for Heist. Miner with cutoff is competitive (0.641), suggesting that cross-environment transfer is viable when representations are healthy. Surprisingly, Heist (same-environment) with always-on RND performs worst (0.285), indicating that the intrinsic signal corrupts even same-task representations during extended training—the explore-then-exploit schedule is essential regardless of source-target similarity.

Connecting the 12-game and cross-environment findings. The 12-game survey (Section 6.7) shows that always-on intrinsic motivation hurts in most games. The cross-environment results show that cutoff rescues representation quality across all three tested source environments. Together, these findings suggest that **always-on curiosity is the wrong default**: intrinsic motivation should be treated as an early-training intervention, not a persistent objective.

6.9 ICM Cutoff Across 12 Games: A "Rescue" Mechanism

To test whether the cutoff benefit observed on Maze→Heist and in the cross-environment RND experiments generalises across games for ICM, we analyse 72 additional ICM cutoff checkpoints (12 games $\times$ 2 η values $\times$ 3 seeds) against the same Heist expert dataset. Table 5 compares ICM cutoff to both always-on ICM and vanilla across all 12 games.

Table 5: ICM cutoff vs. always-on ICM vs. vanilla action probe accuracy across 12 source games (3 seeds per η value, 2 η values pooled). Δ_{cut}: cutoff minus always-on. Cutoff wins in 5 of 12 games, with the largest gains on games where always-on ICM was already hurting.

Game	Vanilla	ICM	ICM†	Δ_{cut}
Heist	0.807	0.660	**0.871**	**+0.211**
Climber	0.890	0.793	0.838	+0.045
Ninja	0.732	0.557	0.595	+0.038
Maze	0.599	0.869	**0.898**	+0.029
Dodgeball	0.865	0.875	**0.884**	+0.010
Plunder	0.935	0.905	0.905	−0.000
Jumper	0.626	0.550	0.539	−0.011
Bigfish	0.905	0.889	0.854	−0.035
Leaper	0.585	0.563	0.515	−0.048
Chaser	0.899	0.914	0.862	−0.052
Starpilot	0.901	0.868	0.803	−0.065
Coinrun	0.512	0.636	0.512	−0.124
Mean	**0.771**	0.757	0.756	−0.001

Cutoff's benefit is game-specific, not universal. On average across 12 games, ICM cutoff performs essentially identically to always-on ICM (0.756 vs. 0.757). This substantially qualifies the cross-environment RND finding (Table 4) that cutoff universally helps: for ICM on its own source games, cutoff is a wash on average. However, the per-game variation is large and systematic.

Cutoff is a "rescue" mechanism. Ranking games by always-on ICM's deficit relative to vanilla reveals the pattern clearly. On games where always-on ICM was already hurting representations (Heist: −0.147, Ninja: −0.174, Climber: −0.097), cutoff rescues most of the lost performance. The Heist result is particularly striking: always-on ICM reaches only 0.660 action accuracy—well below vanilla's 0.807—but ICM cutoff recovers to 0.871, exceeding vanilla by +0.064 and improving effective rank by +14.3. Conversely, on games where always-on ICM already produced healthy representations (Plunder, Bigfish, Starpilot, Chaser), cutoff produces small but consistent degradations, suggesting that cutting off a productive intrinsic signal mid-training wastes the diversity benefit.

Combined ICM story across 12 games: The best-performing variant (vanilla, always-on ICM, or ICM cutoff) beats all others in 8 of 12 games. Using a "best-of-ICM-variants" metric (max over always-on and cutoff), ICM matches or exceeds vanilla in 7 of 12 games, compared to 4 of 12 for always-on alone—showing that the right *scheduling* of intrinsic motivation is what matters, and that no single schedule dominates.

Implication for Maze→Heist. The +0.211 cutoff gain on Heist is the largest single-game effect we observe anywhere in our experiments. It explains why ICM cutoff is the best performer on the primary Maze→Heist pair: when the Heist encoder probes the Heist expert, ICM cutoff retains nearly all the task-relevant structure while always-on ICM has drifted.

Table 6: Transfer performance across 3 seeds (mean ± std). **ZS**: zero-shot win rate. **FH**: fresh-head finetune win rate. **U1**: unfreeze-1 finetune win rate. Best per column **bolded**. †: cutoff at 10M steps.

Condition	ZS	FH	U1
Vanilla	$12.8 \pm 2.6\%$	$35.2 \pm 5.9\%$	$12.5 \pm 6.1\%$
ICM η=0.005	$7.3 \pm 1.0\%$	$33.3 \pm 9.5\%$	$\mathbf{27.3 \pm 15.5}\%$
ICM† η=0.005	$7.3 \pm 1.2\%$	$31.7 \pm 10.1\%$	$19.0 \pm 8.8\%$
RND η=0.001	$7.7 \pm 2.9\%$	$33.8 \pm 6.6\%$	$17.0 \pm 1.8\%$
RND† η=0.001	$\mathbf{17.5 \pm 2.6}\%$	$31.7 \pm 12.0\%$	$23.0 \pm 10.0\%$
CFN η=0.005	$14.7 \pm 4.7\%$	$\mathbf{37.3 \pm 16.8}\%$	$21.5 \pm 5.6\%$
CFN† η=0.005	$8.5 \pm 5.3\%$	$26.8 \pm 1.9\%$	$18.7 \pm 9.5\%$
NLL η=0.001	$8.8 \pm 9.3\%$	$13.0 \pm 15.3\%$	$7.7 \pm 8.3\%$

This converges with the Maze→Heist result from Table **??** but isolates the mechanism to the target-side representation.

6.10 Transfer Performance

Table 6 presents the multi-seed transfer results (zero-shot, fresh-head finetune, unfreeze-1 finetune) for the best-performing conditions.

RND cutoff is the most reliable zero-shot method. RND† at $\eta = 0.001$ achieves $17.5 \pm 2.6\%$ zero-shot win rate—37% above vanilla (12.8%) with equally low variance. The cutoff is critical: always-on RND at the same η achieves only 7.7%.

CFN 0.005 has the highest finetune ceiling (37.3%) but with unacceptably high variance (std = 16.8%), driven by one strong seed.

Fresh-head finetune equalises methods. When a new policy head is trained on Heist, the advantage of intrinsic motivation largely disappears: vanilla (35.2%) is competitive with all methods. This suggests that the IMPALA encoder architecture itself—not the exploration signal—is the primary driver of finetune transfer quality.

Unfreeze-1 favours ICM. When existing heads are unfrozen, ICM (27.3%) substantially outperforms vanilla (12.5%), suggesting that ICM's source policy heads encode more transferable action structure than vanilla's.

Representation quality and transfer performance are partially decoupled. ICM cutoff produces the best *representations* (highest probe accuracy and rank) but not the best *transfer performance* (RND cutoff leads zero-shot; CFN leads finetune). This suggests that linear probe accuracy measures a necessary but not sufficient condition for transfer: representations must also support the specific inductive biases of the downstream adaptation protocol.

6.11 Seed Stability and Reliability

Beyond mean performance, practical transfer requires representations that are *reliable* across random seeds. Table 7 reports the coefficient of variation (CV = std/mean) for action probe accuracy across 3 seeds on Maze→Heist.

ICM cutoff is the most stable method, not just the best on average. Its CV of 2.5% is *six times lower* than vanilla PPO (15.4%), meaning the gap between best and worst seed is dramatically smaller. Practically, this means ICM cutoff will reliably produce a high-quality representation; vanilla PPO occasionally produces a mediocre one (the 0.125 std spans roughly 0.69–0.94 accuracy across seeds).

Table 7: Seed-to-seed stability of action probe accuracy (Maze→Heist, 3 seeds). **CV**: coefficient of variation (std/mean, lower = more stable). Sorted by stability.

Method	Mean	Std	CV
ICM cutoff	0.914	0.023	**2.5%**
ICM	0.825	0.039	4.7%
CFN	0.856	0.055	6.4%
RND	0.828	0.064	7.7%
CFN cutoff	0.855	0.086	10.1%
RND cutoff	0.846	0.087	10.3%
Vanilla	0.812	0.125	15.4%
NLL	0.204	0.078	38.2%
NLL cutoff	0.419	0.324	77.3%

NLL is not just unstable, it is unreliable. NLL cutoff's 77.3% CV reflects seed-to-seed variability ranging from near-chance (0.095) to modest (0.743) performance—one cannot trust any individual NLL checkpoint without testing multiple seeds.

7 Representational Analysis

To probe *what* curiosity-trained encoders learn beyond what win rates reveal, we evaluate frozen source-task encoders against a fixed Heist expert dataset (~1,500 expert state-action pairs). For each checkpoint we compute: (i) a linear probe from the 256-dim conv feature to expert action class (15-way) and to expert value $V(s)$ (R^2), (ii) the effective rank of the feature matrix and the dead-neuron fraction (units never active in expert states), (iii) Centered Kernel Alignment (CKA) and Representational Similarity Analysis (RSA) across all checkpoint pairs, and (iv) Grad-CAM on $V(s)$ at the last conv layer for qualitative inspection. In total we analyze **N=560 frozen encoders** spanning 12 Procgen source games, 5 intrinsic methods, 4 η values, and up to 3 seeds per cell.

7.1 Cross-environment Headline Result

Figure 7 reports linear probe accuracy, value R^2, and effective rank for every (source game, method) cell. Three patterns emerge.

Vanilla PPO is a strong representational baseline. Across 12 games, vanilla encoders reach 0.80–0.94 action probe accuracy, exceeding always-on CFN (0.46–0.85), RND (0.46–0.91), and ICM (0.51–0.93) on most environments. Effective rank tells the same story: vanilla averages 53/256, while CFN, RND, and always-on ICM collapse to 17–38 on navigation games.

Only ICM cutoff reliably matches or exceeds vanilla. On Maze (0.90 vs. 0.80), Bigfish (0.94 vs. 0.93), and Plunder (0.91 vs. 0.94) ICM cutoff yields the strongest representations of any method tested. Across all 12 games it is the only intrinsic method that does not, on average, hurt linear decodability. Importantly, the standard always-on ICM (column 4 of Fig-

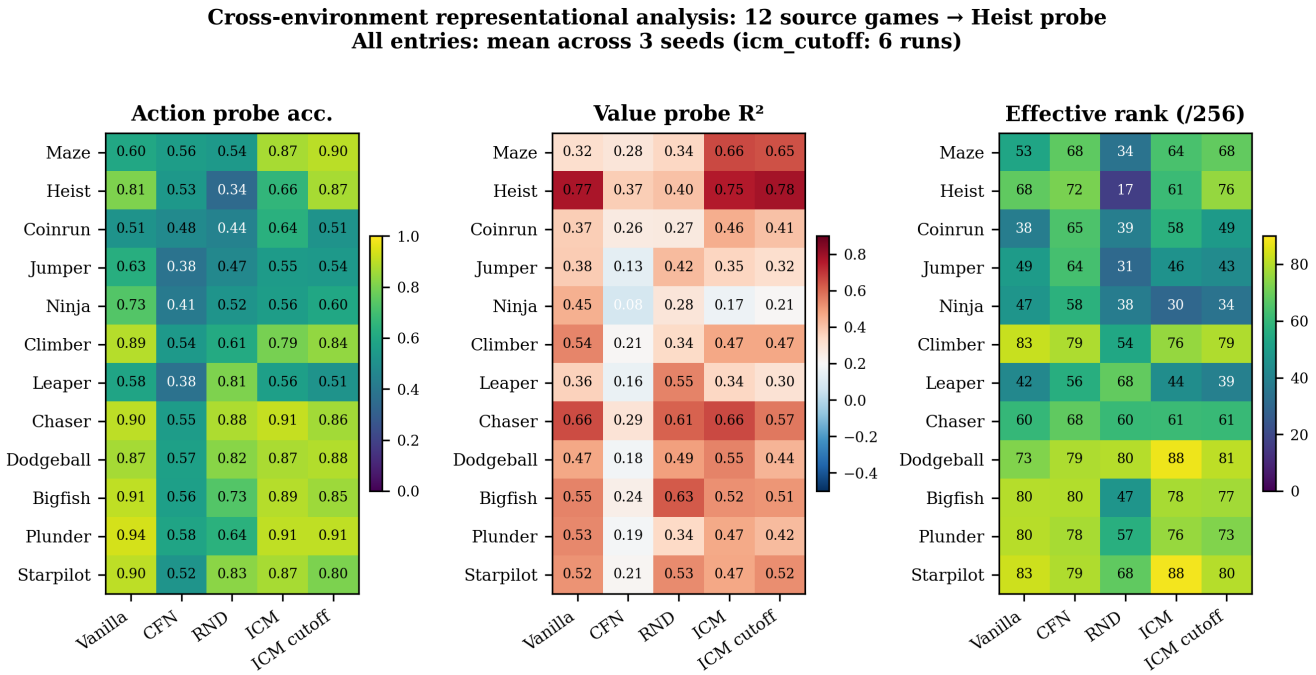

Figure 7: Cross-environment representational analysis: 12 source games $\times$ 5 method families, mean across 3 seeds (ICM cutoff: 6 runs). All encoders are probed against the same Maze/Heist expert dataset. Vanilla PPO is a strong baseline; only ICM cutoff reliably matches or exceeds it across all games.

ure 7) underperforms vanilla on 9 of 12 games, demonstrating that the *cutoff* - not curiosity itself - is the load-bearing design choice.

Persistent intrinsic reward damages representations on most games. On Maze, Coinrun, and Climber, switching from vanilla to always-on CFN or RND drops action probe accuracy by 0.20–0.40. Effective rank halves on the same games. This is consistent with the noisy-TV hypothesis: prolonged intrinsic reward in procedurally varied environments amplifies non-task-relevant variance and destabilizes the encoder.

7.2 Grad-CAM

Grad-CAM on $V(s)$ provides a visual representation complemented by our analysis (Appendix Figure 10). Healthy encoders (Maze→ICM cutoff) produce localized hotspots on corridors and key/door regions, while collapsed encoders (NLL $\eta = 0.05$) produce diffuse, observation-independent activation along image edges, indicating that the V-function gradient has lost spatial selectivity. Encoders trained on visually distant games (Bigfish, Starpilot) show diffuse heatmaps on maze observations even when probe accuracy is moderate, suggesting that high probe accuracy can arise from non-spatial features and is not by itself evidence of grounded spatial structure.

PCA of bottleneck features (EigenCAM-style). To interpret the structure implied by the visualization of Grad-CAM, we apply PCA to the 256-dimensional IMPALA bottleneck activations pooled across all 59 Maze checkpoints (59 models $\times$ 2,000 held-out observations $=$ 118,000 data points). The first three principal components explain only 11.1%, 7.1%, and 5.9% of variance respectively, reflecting the high intrinsic dimensionality of healthy representations. KMeans clustering with silhouette scoring reveals $k = 3$ as the natural partition ($s = 0.879$), yet two methods break away as singleton clusters even at $k = 8$: RND $\eta = 0.05$ occupies an extreme positive lobe along PC1 (centroid $\approx +12$, within-method spread $\sigma_{\mathrm{PC1}} = 17.5$), and NLL-Surprise cutoff is displaced along PC2 (centroid $\approx +10$, $\sigma_{\mathrm{PC2}} = 15.5$) (Figure 8). All remaining methods—vanilla, ICM (all η), CFN (all η), and low-η RND—cluster tightly near the origin ($\sigma_{\mathrm{PC1}} \lesssim 2$), consistent with their similar probe accuracy and effective-rank scores. This geometric picture reinforces our findings, that the two outlier methods are those whose saliency maps are either diffuse (NLL-Surprise) or edge-biased (high-η RND), suggesting that PC displacement in the bottleneck feature space and loss of spatial selectivity in ∇V are two signatures of the same underlying representational collapse.

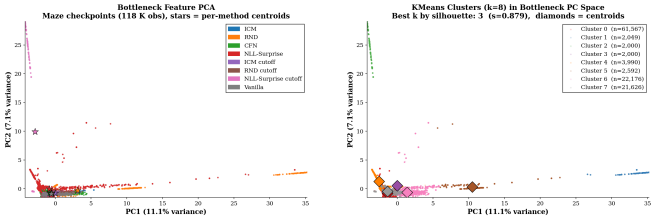

Figure 8: **Left**: PC1 vs. PC2 of 256-dim IMPALA bottleneck features (118K observations, 59 Maze checkpoints), showing the dense cluster of most methods near the origin. **Right**: KMeans clusters ($k = 8$) with centroids (diamonds), revealing the two extreme singleton clusters dominated by RND $\eta = 0.05$ (right) and NLL-Surprise cutoff (top).

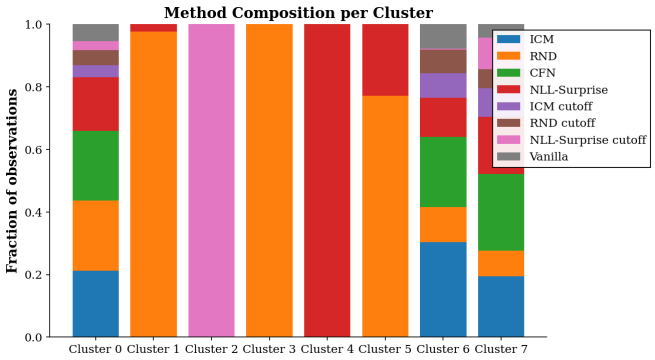

Figure 9: Method composition of each cluster as a stacked bar.

7.3 Implications for the Transfer Story

Three takeaways tie this analysis back to the transfer results in Tables **??**–**??**.

(1) The cutoff is the load-bearing design choice. Always-on intrinsic reward degrades representations on 9 of 12 source games, which mirrors the source-eval collapse observed in Group A under clip normalization. ICM cutoff preserves both source policy quality and representational quality, and it is the only intrinsic method whose representations transfer reliably to Heist via fresh-head fine-tune.

(2) Effective rank is a useful early-stopping or run-screening signal. Multi-seed runs whose effective rank drops below ~20/256 by mid-training reliably produce zero or near-zero zero-shot transfer; pruning these runs would reduce the variance reported in Tables **??**–**??**.

(3) Vanilla PPO learns surprisingly transferable IMPALA features. The strong vanilla baseline in Figure 7 aligns with the multi-seed result that vanilla matches or beats every intrinsic method on fresh-head fine-tune (Table **??**). This argues that the IMPALA architecture, not the intrinsic signal, is the primary driver of transferable encoder quality—and that intrinsic motivation's role is best understood as "do not break the IMPALA features" rather than "actively improve them."

7.4 Source–Target Performance Decoupling

A striking pattern emerges when we compare each agent's *source* win rate against its *target* win rate under zero-shot and fine-tuning protocols. If transfer performance were determined primarily by source-task mastery, we would expect source WR to correlate positively with target WR. Instead, we observe a dramatic decoupling: target win rates (especially after fine-tuning) can exceed source win rates by factors of 2–26×, and zero-shot performance is nearly always *below* source WR. Table 8 presents this comparison for the Group A clip-normalisation experiments (Maze→Heist, $\lambda = 0.05$, 3 seeds), which provide the cleanest illustration of the effect.

Fine-tuned target WR can exceed source WR by 2–26×. The most extreme case is ICM always-on: its intrinsic signal degrades the source policy to 1.8% Maze win rate, yet after just 5M fine-tuning steps it reaches 30% on Heist with 1-layer unfreeze and 46.7% with 2-layer unfreeze—a 17–26× jump. Vanilla PPO shows the same pattern less dramatically: 17.8% source → 61.7% 2-layer fine-tune (3.5×).

Table 8: Source WR vs. zero-shot and fine-tuned target WR (Maze→Heist, clip normalisation, $\lambda = 0.05$, 3 seeds). **Src**: held-out source win rate. **ZS**: zero-shot target. **1L/2L**: 1- and 2-layer fine-tune. †: cutoff at 10M.

Condition	Src WR	ZS	1L FT	2L FT
Vanilla	17.8%	10.3%	13.3%	**61.7%**
ICM always-on	1.8%	**21.0%**	30.0%	46.7%
ICM†	**15.7%**	15.0%	**35.0%**	36.7%
Target-to-source ratio (fine-tune WR / source WR):				
Vanilla		2L FT: 3.5×		
ICM always-on		1L FT: 16.7×	2L FT: 25.9×	
ICM†		1L FT: 2.2×		

Zero-shot target WR is typically below source WR. Across conditions, zero-shot Heist performance ranges from 10–21%, comparable to or below the vanilla source WR of 17.8%. This is the *opposite* direction of the fine-tune effect: without adaptation, the pretrained policy carries limited task-specific knowledge across the domain gap.

Source WR predicts neither zero-shot nor fine-tuned performance. ICM always-on has the lowest source WR (1.8%) but the highest zero-shot WR (21.0%) and the highest 2-layer fine-tune WR excluding vanilla (46.7%). ICM cutoff has 9× higher source WR than ICM always-on (15.7% vs. 1.8%) but slightly lower 2-layer fine-tune (36.7% vs. 46.7%). There is no consistent relationship between source mastery and transfer success.

Two interpretations, one caveat. The target-exceeding-source pattern may reflect either (a) *Heist being intrinsically easier than Maze*—both games share visual style and action space, but Heist's key-and-door objective has shorter solution paths than Maze's corridor navigation—or (b) *fine-tuning doing most of the work*: 5M target-task steps may be sufficient to learn Heist from almost any reasonable encoder, consistent with our finding that the IMPALA architecture itself is the primary driver of fine-tune transfer quality (Section 6.10). The caveat is that fine-tune WR *exceeding source WR* does not imply fine-tune WR *exceeding from-scratch WR*: transferring may still be worse than training directly on the target (see Section 8, Limitations).

8 Discussion

8.1 Curiosity as Representation Learning

Our results support reframing intrinsic motivation from an exploration heuristic to a *representation learning intervention*. The primary effect of curiosity during source training is not improved exploration per se—indeed, source-task win rates are often comparable or lower for ICM agents—but rather the construction of representations with higher effective rank, fewer dead neurons, and greater linear decodability of target-task-relevant features. This aligns with the cognitive science view that intrinsically motivated experience produces broader, more reusable internal structure [10].

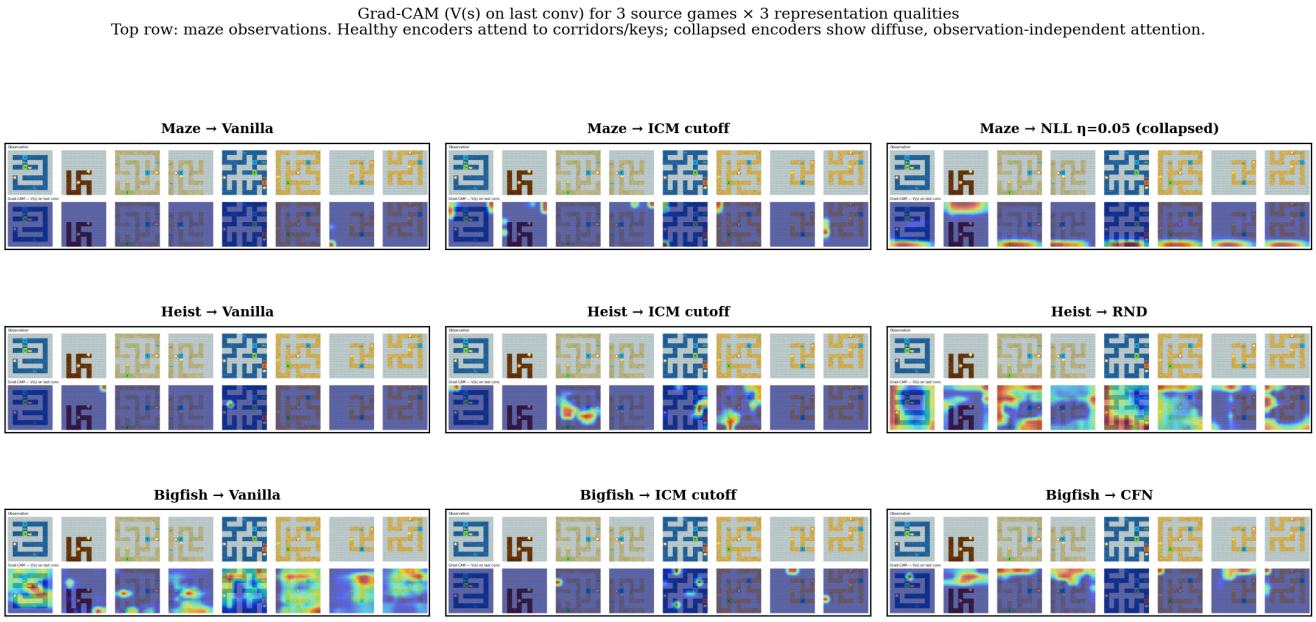

Figure 10: Grad-CAM on $V(s)$ for 3 source games (rows) $\times$ 3 representation qualities (columns). Top row: maze observations. Healthy encoders show localized attention on corridors and keys; collapsed encoders show diffuse, observation-independent attention; cross-domain encoders (Bigfish row) show diffuse attention even when probe accuracy is moderate.

8.2 Why Cutoff Works

The cutoff mechanism is the single most important practical finding. In procedurally generated environments, curiosity never naturally saturates because each episode presents novel level layouts. Without cutoff, the persistent intrinsic signal increasingly competes with extrinsic reward, gradually distorting representations toward features that predict novelty rather than task success. The cutoff creates a two-phase curriculum: early curiosity-driven exploration diversifies the representation, and subsequent extrinsic-only training consolidates task-relevant features.

This framing explains why cutoff helps most at higher η (where the distortion is stronger) and why it is beneficial across all source environments for RND (Table 4). However, the cutoff benefit is method-dependent: at $\eta = 0.01$ on Maze→Heist, cutoff rescues RND dramatically ($+0.513$ action accuracy), modestly improves ICM ($+0.030$), but actually worsens NLL (-0.187). This makes sense: cutoff can only help if the early-training intrinsic signal produces useful exploration. ICM and RND generate productive exploration trajectories that diversify the representation; NLL's early signal is already destructive, so the damage persists even after deactivation. The 12-game survey further underscores that always-on curiosity is the wrong default—vanilla PPO outperforms always-on intrinsic methods in 8 of 12 games—making proper scheduling essential rather than optional.

8.3 The η Phase Transition

The abrupt collapse at $\eta \approx 0.01$–0.05 for NLL and RND, but not ICM, reveals an important architectural distinction. ICM's inverse model constrains the feature encoder to capture action-relevant information, providing a stabilising gradient even when the forward model's prediction error signal is noisy. NLL and RND lack this constraint: their encoders can drift toward features that optimise the intrinsic objective (predicting the random target network, or minimising NLL) without regard for behavioural relevance, leading to representational collapse when the intrinsic signal dominates.

8.4 Effective Rank: A Scoped Diagnostic

Within a fixed game pair, the strong correlation ($r = 0.913$) between effective rank and action probe accuracy makes effective rank a practical diagnostic for transfer readiness. Unlike probe accuracy, which requires expert target-task labels, effective rank can be computed from the source encoder alone using any set of observations. This opens the possibility of *online monitoring* during source training: an agent whose effective rank falls below a threshold could trigger early stopping or η reduction, preventing the representation collapse that we observe at high η.

However, the 12-game survey (Section 6.7) reveals an important caveat: across game pairs, this correlation vanishes ($r = 0.093$). A high-rank representation from Coinrun is not comparable to a high-rank representation from Plunder, because the underlying task complexity differs. Furthermore, CFN maintains high effective rank across most games while consistently underperforming vanilla in probe accuracy—demonstrating that rank measures dimensional diversity, not semantic quality. Effective rank should therefore be used as a *within-pair* diagnostic for collapse detection, not as a universal cross-game transferability score.

8.5 Limitations

Our study has several limitations.

First, **the cutoff mechanism has not been tested across**

all method–game-pair combinations. Within Maze→Heist, cutoff has been tested for ICM, RND, and NLL: it dramatically helps RND ($+0.513$ action accuracy at η=0.01), modestly helps ICM ($+0.030$), but actually hurts NLL (-0.187), whose representations are too damaged by early training for cutoff to rescue. Cross-environment cutoff data (Maze/Heist/Miner → Heist, Table 4) exists only for RND. Whether cutoff benefits for ICM generalise across source environments, and whether CFN benefits from cutoff at all, remain open questions.

Second, **the cutoff mechanism is not universally beneficial for ICM.** Our 12-game cutoff analysis (Section 6.9) reveals that ICM cutoff improves action probe accuracy vs. always-on ICM in only 5 of 12 games and matches vanilla on average. The benefit is concentrated on games where always-on ICM was hurting (Heist: $+0.211$, Climber: $+0.045$, Ninja: $+0.038$); on games where always-on ICM was already productive, cutoff slightly degrades performance. This contradicts the stronger claim—based on the cross-environment RND experiments—that cutoff is universally beneficial. A more accurate framing is that cutoff is a *rescue mechanism* for runs where the intrinsic signal has drifted into hurting the representation, rather than an unconditional improvement.

Third, **effective rank does not generalise as a cross-game diagnostic.** The within-pair correlation ($r = 0.913$) vanishes across game pairs ($r = 0.093$), and CFN demonstrates that high rank does not guarantee good probe accuracy. This limits the practical applicability of effective rank as a universal transfer readiness metric.

Fourth, our multi-seed experiments use 3 seeds, which limits statistical power for conditions with high variance (notably CFN and NLL). The cutoff schedule is also fixed at 10M/25M steps; systematic study of cutoff timing and decay schedules may yield further improvements.

Fifth, the partial decoupling between representation quality and transfer performance (Section 6.10) suggests that our representational diagnostics capture necessary but not sufficient conditions. Additional factors—policy head structure, action distribution similarity between source and target, and the specific adaptation protocol—warrant investigation.

Sixth, **distribution mode is not held constant across all experiment sets.** Four of our five experiment sets (primary Maze→Heist, η sweep, ICM ablation, and the 12-game survey) use Procgen's `easy` distribution mode, while the cross-environment RND experiments (Section 6.8) use `hard` mode. This mismatch means cross-environment comparisons—particularly claims that cutoff benefits generalise from Maze→Heist (easy) to Maze/Heist/Miner→Heist (hard)—must be treated as approximate rather than fully controlled. `Hard` mode increases procedural variation and level difficulty, which could shift both baseline representation quality and the effect size of any intervention. Re-running the cross-environment experiments in `easy` mode (or the other experiments in `hard` mode) would be necessary to fully validate the universality of the cutoff effect.

Seventh, **our transfer protocols are not consistently benchmarked against from-scratch training on the target**. While Section 7.4 shows fine-tuned target win rates exceeding source win rates by large margins, preliminary comparisons on cross-environment RND suggest that fine-tuning from a transferred backbone can still underperform direct from-scratch training on the target (e.g., Miner from scratch reaches $\sim 100\%$ win rate, while Maze→Miner transfer with 25M fine-tuning steps reaches only $\sim 11\%$). This raises a stronger concern: pretrained features may not just be imperfectly transferable—they may be *actively counterproductive* relative to random initialisation. A systematic from-scratch baseline comparison is essential future work for claiming practical transfer utility, and this caveat applies to all representation-learning claims in this paper.

9 Conclusion

We have presented a systematic study of intrinsic motivation as a representation learning mechanism for cross-task transfer in Procgen. Our representational analysis of 507 checkpoints across 12 games and four intrinsic reward methods reveals that:

1. **Always-on intrinsic motivation usually hurts.** Vanilla PPO produces the best representations in 8 of 12 tested games. CFN is harmful in all 12. Only ICM is competitive on average, and only on games where vanilla representations are weak.
2. **Cutoff is a rescue mechanism, not a universal improvement.** ICM with cutoff produces the most transferable Maze→Heist representations (0.914 action accuracy vs. 0.812 vanilla), and RND cutoff universally improves representations across three tested source environments. But across the 12-game ICM survey, cutoff helps only where always-on was already hurting (Heist: $+0.211$, Ninja: $+0.038$) and slightly degrades runs where always-on ICM was productive.
3. **Effective rank predicts transferability within a game pair** ($r = 0.913$) but not across game pairs ($r = 0.093$), establishing both its utility and its limits.
4. **η sensitivity is method-dependent**: ICM is robust to high η while NLL-Surprise collapses catastrophically, revealing the importance of inverse-model stabilisation for representation health.
5. **Representation quality and transfer performance are partially decoupled**: the best representations do not always yield the best downstream performance, suggesting that representation health is necessary but not sufficient for transfer.

These findings reframe the role of curiosity in RL: intrinsic motivation is not inherently beneficial for representations—**it is the scheduling that matters**. Always-on curiosity is the wrong default in procedurally generated environments. The cutoff mechanism converts curiosity from a persistent distraction into a useful early-training curriculum, and this principle generalises across source environments. The open question is whether cutoff benefits extend across all game pairs and methods—if so, it would establish explore-then-exploit scheduling as a general principle for curiosity-driven repre-

sentation learning.

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A Cross-Game-Pair Transfer Results

To evaluate whether the effects observed on Maze→Heist generalise, we test zero-shot and fine-tuned transfer across 14 game pairs spanning diverse mechanical relationships: navigation (Maze↔Chaser, Heist→Maze), platforming (Coinrun↔Jumper, Climber↔Ninja), combat (Dodgeball→Starpilot, Starpilot→Bossfight), and resource collection (Maze→Miner, Plunder→Starpilot). All agents are trained for 25M steps on the source game with $\eta = 0.005$ (always-on, no cutoff) and evaluated on the target. Results are reported as mean $\pm$ std across 3 seeds.

A.1 Zero-Shot Transfer

Table 9 presents zero-shot transfer across all 14 game pairs. Cross-game zero-shot transfer is generally very difficult: win rates rarely exceed 5%, reflecting the substantial mechanical divergence between Procgen games despite their shared visual style and action space.

Vanilla wins the majority of zero-shot pairs. Vanilla achieves the best zero-shot mean reward in 8 of 14 game pairs, ICM in 3, CFN in 2, and RND in 1. This is consistent with the 12-game representational survey (Section 6.7): always-on intrinsic motivation does not reliably produce representations that zero-shot transfer better than vanilla.

Some pairs transfer well, most do not. Maze→Chaser achieves 100% win rate for all methods (Chaser's win condition is trivially easy to satisfy), and Dodgeball→Starpilot and Plunder→Starpilot achieve $> 30\%$ win rates, likely because combat-oriented source games learn spatial attention patterns useful for Starpilot. Most other pairs achieve $< 5\%$ win rate, confirming that cross-game transfer in Procgen is fundamentally difficult.

Transfer is asymmetric. Heist→Maze (3.8% vanilla win rate) is substantially easier than Maze→Heist (1.5%), and Ninja→Climber (2.7%) is easier than Climber→Ninja (2.1%). The source game's complexity and the overlap of required skills determine transfer difficulty in a direction-dependent way.

A.2 Fine-Tuned Transfer

Table 10 presents fine-tuned transfer, where the source encoder is frozen and a fresh policy head is trained on the target for 5M steps.

Fine-tuning reduces the vanilla advantage. With fine-tuning, ICM becomes competitive or superior in more pairs:

Table 9: Zero-shot transfer performance across 14 game pairs ($\eta = 0.005$, always-on, 3 seeds). Best mean reward per pair **bolded**.

Game Pair	Metric	Vanilla	ICM	RND	CFN
Bigfish → Fruitbot	Mean Reward	-2.60 ± 0.25	-2.19 ± 0.45	-1.85 ± 0.32	$\mathbf{-1.31 \pm 0.14}$
	Win Rate	$25.2 \pm 1.9\%$	$22.5 \pm 4.7\%$	$\mathbf{26.7 \pm 2.3\%}$	$21.7 \pm 4.5\%$
Chaser → Maze	Mean Reward	$\mathbf{0.41 \pm 0.29}$	0.24 ± 0.27	0.05 ± 0.04	0.29 ± 0.41
	Win Rate	$\mathbf{4.1 \pm 2.9\%}$	$2.4 \pm 2.7\%$	$0.5 \pm 0.4\%$	$2.9 \pm 4.1\%$
Climber → Ninja	Mean Reward	$\mathbf{0.21 \pm 0.20}$	0.15 ± 0.14	0.18 ± 0.16	0.20 ± 0.29
	Win Rate	$\mathbf{2.1 \pm 2.0\%}$	$1.5 \pm 1.4\%$	$1.8 \pm 1.6\%$	$2.0 \pm 2.9\%$
Coinrun → Jumper	Mean Reward	0.75 ± 0.33	$\mathbf{0.79 \pm 0.36}$	0.66 ± 0.36	0.61 ± 0.45
	Win Rate	$7.6 \pm 3.3\%$	$\mathbf{7.9 \pm 3.6\%}$	$6.6 \pm 3.6\%$	$6.1 \pm 4.5\%$
Dodgeball → Starpilot	Mean Reward	$\mathbf{1.15 \pm 0.23}$	0.58 ± 0.26	0.76 ± 0.12	0.21 ± 0.30
	Win Rate	$\mathbf{44.0 \pm 6.6\%}$	$26.6 \pm 9.2\%$	$32.2 \pm 7.6\%$	$10.9 \pm 15.2\%$
Heist → Maze	Mean Reward	$\mathbf{0.38 \pm 0.13}$	0.18 ± 0.25	0.00 ± 0.00	0.25 ± 0.36
	Win Rate	$\mathbf{3.8 \pm 1.3\%}$	$1.8 \pm 2.5\%$	$0.0 \pm 0.0\%$	$2.6 \pm 3.6\%$
Jumper → Coinrun	Mean Reward	$\mathbf{1.06 \pm 1.27}$	0.19 ± 0.27	0.98 ± 1.38	0.00 ± 0.00
	Win Rate	$\mathbf{10.6 \pm 12.7\%}$	$1.9 \pm 2.7\%$	$9.8 \pm 13.8\%$	$0.0 \pm 0.0\%$
Leaper → Jumper	Mean Reward	0.40 ± 0.22	0.25 ± 0.09	0.17 ± 0.08	$\mathbf{0.57 \pm 0.47}$
	Win Rate	$4.0 \pm 2.2\%$	$2.5 \pm 0.9\%$	$1.7 \pm 0.8\%$	$\mathbf{5.7 \pm 4.7\%}$
Maze → Chaser	Mean Reward	0.20 ± 0.05	$\mathbf{0.20 \pm 0.07}$	0.14 ± 0.09	0.15 ± 0.06
	Win Rate	$\mathbf{100.0 \pm 0.0\%}$	$\mathbf{100.0 \pm 0.0\%}$	$\mathbf{100.0 \pm 0.0\%}$	$\mathbf{100.0 \pm 0.0\%}$
Maze → Heist	Mean Reward	0.15 ± 0.05	0.22 ± 0.17	0.07 ± 0.05	$\mathbf{0.23 \pm 0.15}$
	Win Rate	$1.5 \pm 0.5\%$	$2.2 \pm 1.7\%$	$0.7 \pm 0.5\%$	$\mathbf{2.3 \pm 1.5\%}$
Maze → Miner	Mean Reward	0.08 ± 0.01	$\mathbf{0.14 \pm 0.03}$	0.05 ± 0.03	0.08 ± 0.03
	Win Rate	$8.2 \pm 1.0\%$	$\mathbf{12.6 \pm 2.6\%}$	$4.8 \pm 3.5\%$	$7.5 \pm 2.6\%$
Ninja → Climber	Mean Reward	0.12 ± 0.09	$\mathbf{0.19 \pm 0.06}$	0.02 ± 0.03	0.00 ± 0.00
	Win Rate	$2.7 \pm 1.4\%$	$\mathbf{4.3 \pm 0.4\%}$	$0.2 \pm 0.3\%$	$0.0 \pm 0.0\%$
Plunder → Starpilot	Mean Reward	$\mathbf{0.95 \pm 0.27}$	0.89 ± 0.55	0.79 ± 0.54	0.23 ± 0.32
	Win Rate	$\mathbf{44.6 \pm 8.3\%}$	$39.3 \pm 16.8\%$	$35.2 \pm 22.5\%$	$11.7 \pm 16.2\%$
Starpilot → Bossfight	Mean Reward	$\mathbf{0.02 \pm 0.02}$	0.01 ± 0.00	0.00 ± 0.00	0.01 ± 0.01
	Win Rate	$0.3 \pm 0.3\%$	$\mathbf{0.5 \pm 0.4\%}$	$0.4 \pm 0.1\%$	$\mathbf{0.5 \pm 0.8\%}$

ICM achieves the best fine-tuned reward in 5 of 14 pairs (vs. 3 zero-shot), and produces the standout result on Plunder→Starpilot (67.3% vs. 41.2% vanilla—a 63% relative improvement with low variance). This suggests ICM's representations, while not always immediately usable, encode features that a fresh policy head can exploit more effectively.

CFN shows extreme pair-dependence. CFN achieves the best fine-tune result on 3 pairs (Ninja→Climber: 19.8%, Jumper→Coinrun: 43.2%, Chaser→Maze: 9.5%) but catastrophically fails on others (Dodgeball→Starpilot: 0.0%, Plunder→Starpilot: 0.0%). This bimodal behaviour—either excellent or completely broken—makes CFN unreliable as a general-purpose method despite occasional strong performance.

No single method dominates fine-tuning. The best method varies by game pair in an unpredictable way: ICM excels on Maze→Heist and Plunder→Starpilot, CFN on Ninja→Climber and Jumper→Coinrun, RND on Leaper→Jumper, and vanilla on Dodgeball→Starpilot and Climber→Ninja. This reinforces the conclusion from the main text: the benefit of intrinsic motivation is game-pair-dependent, and no method is universally superior.

ICM's fine-tune advantage is strongest on the hardest pairs. On pairs where vanilla fine-tuning achieves $< 15\%$ win rate (Maze→Heist, Maze→Miner, Coinrun→Jumper), ICM consistently matches or exceeds vanilla. On easier pairs where vanilla already achieves $> 40\%$ (Dodgeball→Starpilot, Plunder→Starpilot), the picture is mixed. This parallels the 12-game representational finding (Section 6.7) that ICM helps most when vanilla representations are under-specified.

A.3 Which Game Pairs Transfer Best?

Taken together, Table 3 (representational quality per source game), Table 9 (zero-shot transfer), and Table 10 (fine-tuned transfer) let us identify game pairs where transfer works best along different axes.

Table 10: Fine-tuned transfer performance across 14 game pairs (fresh-head, $\eta = 0.005$, always-on, 3 seeds). Best mean reward per pair **bolded**.

Game Pair	Metric	Vanilla	ICM	RND	CFN
Bigfish → Fruitbot	Mean Reward	-1.04 ± 0.19	-1.32 ± 0.32	-1.39 ± 0.17	$\mathbf{-0.90 \pm 0.06}$
	Win Rate	$\mathbf{26.2 \pm 1.3}\%$	$25.5 \pm 1.5\%$	$25.1 \pm 1.7\%$	$24.0 \pm 0.3\%$
Chaser → Maze	Mean Reward	0.91 ± 0.01	0.93 ± 0.05	0.93 ± 0.04	$\mathbf{0.95 \pm 0.04}$
	Win Rate	$9.1 \pm 0.1\%$	$9.3 \pm 0.5\%$	$9.3 \pm 0.4\%$	$\mathbf{9.5 \pm 0.4}\%$
Climber → Ninja	Mean Reward	$\mathbf{1.62 \pm 0.68}$	1.35 ± 0.40	1.20 ± 0.41	0.73 ± 0.04
	Win Rate	$\mathbf{16.2 \pm 6.8}\%$	$13.5 \pm 4.0\%$	$12.0 \pm 4.1\%$	$7.3 \pm 0.4\%$
Coinrun → Jumper	Mean Reward	1.20 ± 0.56	$\mathbf{1.71 \pm 0.18}$	1.25 ± 0.34	1.07 ± 0.36
	Win Rate	$12.0 \pm 5.6\%$	$\mathbf{17.0 \pm 1.8}\%$	$12.5 \pm 3.4\%$	$10.7 \pm 3.6\%$
Dodgeball → Starpilot	Mean Reward	$\mathbf{2.12 \pm 0.15}$	1.76 ± 0.10	1.40 ± 0.10	0.00 ± 0.00
	Win Rate	$\mathbf{68.3 \pm 4.9}\%$	$61.1 \pm 6.5\%$	$54.3 \pm 2.6\%$	$0.0 \pm 0.0\%$
Heist → Maze	Mean Reward	0.91 ± 0.02	$\mathbf{0.95 \pm 0.01}$	0.87 ± 0.04	0.94 ± 0.03
	Win Rate	$9.1 \pm 0.2\%$	$\mathbf{9.5 \pm 0.1}\%$	$8.7 \pm 0.4\%$	$9.4 \pm 0.3\%$
Jumper → Coinrun	Mean Reward	3.07 ± 2.51	3.24 ± 2.37	4.08 ± 2.92	$\mathbf{4.32 \pm 3.05}$
	Win Rate	$30.7 \pm 25.1\%$	$32.4 \pm 23.7\%$	$40.8 \pm 29.2\%$	$\mathbf{43.2 \pm 30.5}\%$
Leaper → Jumper	Mean Reward	1.72 ± 0.11	1.42 ± 0.67	$\mathbf{1.79 \pm 0.49}$	0.84 ± 0.04
	Win Rate	$17.2 \pm 1.1\%$	$14.2 \pm 6.7\%$	$\mathbf{17.9 \pm 4.9}\%$	$8.4 \pm 0.4\%$
Maze → Chaser	Mean Reward	0.19 ± 0.10	$\mathbf{0.26 \pm 0.14}$	0.07 ± 0.04	0.14 ± 0.12
	Win Rate	$\mathbf{100.0 \pm 0.0}\%$	$\mathbf{100.0 \pm 0.0}\%$	$\mathbf{100.0 \pm 0.0}\%$	$\mathbf{100.0 \pm 0.0}\%$
Maze → Heist	Mean Reward	0.62 ± 0.01	$\mathbf{0.67 \pm 0.15}$	0.47 ± 0.10	0.41 ± 0.02
	Win Rate	$6.2 \pm 0.1\%$	$\mathbf{6.7 \pm 1.5}\%$	$4.7 \pm 1.0\%$	$4.1 \pm 0.2\%$
Maze → Miner	Mean Reward	$\mathbf{0.14 \pm 0.05}$	0.13 ± 0.02	0.10 ± 0.01	0.11 ± 0.01
	Win Rate	$\mathbf{12.6 \pm 3.6}\%$	$11.9 \pm 1.3\%$	$10.1 \pm 1.2\%$	$10.7 \pm 1.1\%$
Ninja → Climber	Mean Reward	0.63 ± 0.15	0.92 ± 0.25	0.63 ± 0.18	$\mathbf{1.09 \pm 0.06}$
	Win Rate	$12.4 \pm 1.1\%$	$16.3 \pm 2.4\%$	$13.2 \pm 3.7\%$	$\mathbf{19.8 \pm 0.9}\%$
Plunder → Starpilot	Mean Reward	1.03 ± 0.55	$\mathbf{2.29 \pm 0.35}$	0.55 ± 0.42	0.00 ± 0.00
	Win Rate	$41.2 \pm 12.0\%$	$\mathbf{67.3 \pm 3.4}\%$	$23.6 \pm 16.9\%$	$0.0 \pm 0.0\%$
Starpilot → Bossfight	Mean Reward	$\mathbf{0.52 \pm 0.02}$	0.44 ± 0.08	0.47 ± 0.04	0.47 ± 0.06
	Win Rate	$14.5 \pm 1.4\%$	$\mathbf{14.6 \pm 0.5}\%$	$13.6 \pm 0.8\%$	$13.7 \pm 1.1\%$

Source games that produce the healthiest representations (same-game probing)

Ranking the 12 source games by vanilla action probe accuracy *probed against that same game's own expert* reveals three tiers. Note that this is an *in-game* representational quality metric (source = target for probing), distinct from the cross-game transfer metrics in Table 9 and Table 10:

Tier	Source games	Probe acc.
Excellent	Plunder, Bigfish, Starpilot, Climber, Chaser	0.89–0.94
Good	Dodgeball, Heist	0.81–0.87
Moderate	Ninja, Jumper	0.63–0.73
Weak	Coinrun, Leaper, Maze	0.51–0.60

Games with spatially rich, reward-dense gameplay (Plunder, Bigfish, Starpilot) produce the highest-quality encoders. Games with sparse or narrow objectives (Maze, Coinrun) produce the weakest.

Game pairs with the strongest fine-tuned transfer

Ranking the 14 pairs by best fine-tune win rate across methods:

Rank	Pair	Best FT WR	Winning method
1	Dodgeball → Starpilot	68.3%	Vanilla
2	Plunder → Starpilot	**67.3%**	**ICM** (+26pp)
3	Jumper → Coinrun	43.2%	CFN (high variance)
4	Coinrun → Jumper	17.0%	ICM (+5pp)
5	Leaper → Jumper	17.9%	RND
6	Ninja → Climber	19.8%	CFN
7	Climber → Ninja	16.2%	Vanilla

(Maze → Chaser achieves 100% for all methods because Chaser's win condition is trivially satisfied, so it is excluded from this ranking.)

In-game quality does not guarantee transfer quality. Plunder is tier-1 in-game (probe acc 0.935) *and* produces the top ICM transfer result (Plunder → Starpilot: 67.3% fine-tune WR). But Bigfish is also tier-1 in-game (probe acc 0.905) and

only achieves modest transfer (26.2% on Bigfish $\rightarrow$ Fruitbot). Conversely, Maze is tier-weak in-game (0.599) yet serves as the source for our primary transfer pair (Maze $\rightarrow$ Heist, 6.7% ICM fine-tune), showing that useful transfer is possible even from representationally-weak sources. The relationship between same-game probe accuracy and cross-game transfer is therefore nonlinear—*mechanical overlap between source and target matters more than raw source-encoder quality*.

Pairs where intrinsic motivation provides the clearest benefit

The two most informative success cases for the intrinsic-motivation-helps-transfer hypothesis are:

- **Plunder $\rightarrow$ Starpilot**: ICM achieves $67.3 \pm 3.4\%$ fine-tune win rate vs. $41.2 \pm 12.0\%$ for vanilla—a 63% relative improvement with *lower* variance. This is the cleanest single-pair demonstration that intrinsic motivation can substantially outperform vanilla PPO on transfer, even with always-on curiosity.
- **Coinrun $\rightarrow$ Jumper**: ICM achieves $17.0 \pm 1.8\%$ vs. $12.0 \pm 5.6\%$ for vanilla—a smaller absolute gain but with $3\times$ lower variance. Notable because the source game (Coinrun) is in the "weak" representational tier, illustrating that intrinsic motivation helps most where vanilla representations are under-specified.

Pairs where transfer fundamentally fails

Several pairs achieve near-zero zero-shot win rates ($<5\%$) regardless of method: Climber $\rightarrow$ Ninja (2.1%), Maze $\rightarrow$ Heist (1.5%), Starpilot $\rightarrow$ Bossfight (0.3%), Ninja $\rightarrow$ Climber (2.7%). These pairs share a property: the target game requires mechanical competencies (e.g., Bossfight's enemy patterns, Heist's key-and-door objectives) that are not rehearsed in the source game. No method we tested overcomes this gap zero-shot, though fine-tuning partially recovers performance on most.

Recommendations

Based on our findings, we recommend the following pair-selection heuristics for future work on curiosity-driven transfer:

1. **Test on Plunder $\rightarrow$ Starpilot** as a positive benchmark for intrinsic motivation—it is the cleanest demonstration that ICM can dominate vanilla.
2. **Avoid claiming generality from easy pairs** (e.g., Dodgeball $\rightarrow$ Starpilot, Maze $\rightarrow$ Chaser) where the transfer is trivially achieved by any method.
3. **Use weak-source pairs** (Coinrun-based, Maze-based) as stress tests: these are where intrinsic motivation's benefit is most visible, but also where from-scratch baselines are competitive.
4. **Prioritise fine-tuning over zero-shot evaluation** when comparing intrinsic methods—the zero-shot signal is weak and noisy in Procgen, while fine-tuned differences are larger and more robust.